

Spatio-Temporal Crime Hotspot Detection and Pattern Evolution Using Unsupervised Learning

CS719 – Data Science Project

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— Hemin Rohitbhai Shah

ABSTRACT

This project presents a Spatio-temporal analysis of urban crime data using unsupervised learning technique. The main purpose of this project is to identify crime hotspots and understand how these hotspots evolve over time. This project uses the Chicago Crime Dataset, which contains millions of crime records with geographic and temporal information.

Spatial hotspots are detected using the DBSCAN clustering algorithm with haversine distance, which helps to identify dense regions of crime while handling noise effectively. The data is divided into quarterly time windows to capture meaningful temporal changes without excessive or external noise. For each quarter, clusters are generated and summarized using their centroids.

For analysis of hotspot evolution, clusters from consecutive (i.e. previous and current) quarters are compared using centroid distance. Based on this comparison, hotspots are classified as persisted, emerged, or disappeared. This provides a simple and interpretable way to study the stability and dynamics of crime patterns over time.

Finally, results show that most crime hotspots remain stable across time approximately 70 to 85 hotspots, with persistence rates in between 50% to 75%. Although, some areas exhibit more changes in particular years, indicating evolving crime patterns. An interactive Power BI dashboard is developed to visualize spatial distribution, temporal trends, and hotspot evolution, supporting interactive exploration of the results.

Overall, this project demonstrates how unsupervised learning can be used to reveal meaningful Spatio-temporal patterns in large-scale crime data without relying on labelled outcomes.

Table of Contents

1. Introduction	1
1.1. Problem Description	1
1.2. Problem Specification	1
1.3. Goal of the Project	1
1.4. Usefulness of the Project	2
1.5. Structure of the Report	2
2. Background	3
2.1. Previous work.....	3
2.2. Related Research.....	3
2.3. Existing Techniques	4
2.4. Summary of Techniques.....	5
3. Approach	5
3.1. Data Source	5
3.2. Data Preprocessing	6
3.3. Spatial Hotspot Detection	6
3.4. Temporal Segmentation	7
3.5. Hotspot Evolution Method	8
3.6. Example Workflow	9
4. Implementation.....	10
4.1. Tools and Technologies	10
4.2. System Design	11
4.3. Key Code Snippets.....	12
5. Results	13
5.1. Dataset Overview	13
5.2. Clustering Results.....	14
5.3. Hotspot Evolution Results	14
5.4. Multi-Year Analysis	15
5.5. Dashboard Visualization.....	16
6. Conclusion	18
7. Future Work	18
8. References.....	19

1. Introduction

1.1. Problem Description

Crime is not randomly distributed across a city as it usually concentrates in specific geographic areas and varies over time due to factors such as migration or changing population, city or infrastructure development, and due to new policy changes. Understanding where crime occurs and how it changes over time is important for improving public safety and supporting data-driven decision-making.

Traditional crime analysis methods often focus on identifying static hotspots or predicting individual crime events like burglary or theft. Nowadays, these approaches may not fully capture how crime patterns evolve over time. In many real-world cases, it is more useful to understand the stability (i.e. persistence) and changes (i.e. evolving) in crime hotspots instead of focusing only on prediction.

1.2. Problem Specification

This project addresses an unsupervised learning problem in the domain of spatio-temporal data analysis problem to large-scale crime data.

The input to the system consists of crime incident records obtained from the Chicago Crime Dataset. Each record includes spatial and temporal attributes, specifically the latitude and longitude of the incident and the corresponding timestamp (i.e. dates with quarter). These attributes are used to represent the location and time of each crime event.

The output of the system includes spatial clusters which are representing crime hotspots, along with summary statistics, for example, cluster size and noise percentage. In addition, the system produces evolution labels that describe how hotspots change between consecutive time periods. Each hotspot is classified as persisted, emerged, or disappeared based on its spatial relationship with clusters in adjacent time windows. These outputs provide a structured representation of crime patterns, enabling further analysis and visualization of how hotspots evolve across time.

1.3. Goal of the Project

The main objective of this project is to identify spatial crime hotspots and analyse how these hotspots evolve over time using unsupervised learning techniques.

This is achieved by applying density-based clustering to detect regions with high concentrations of crime incidents (area with high volume of crime occurrence) and then tracking how these regions change across consecutive time periods. The goal is not to predict individual crime events, but to understand the stability and dynamics of crime patterns at a regional level. The project also aims to provide an interpretable and scalable framework that can be applied to any large-scale datasets without requiring labelled data. By focusing on pattern discovery, the approach enables meaningful insights into long-term crime behavior.

1.4. Usefulness of the Project

This project is useful for several reasons:

It provides an interpretable framework to analyse how crime hotspots evolve over time without requiring labelled data and helps to identify stable and high-risk areas where crime occurs consistently. It also enables understanding of dynamic changes in crime, for instance, emerging or disappearing hotspots. Finally, it offers a scalable approach that can be applied to large real-world datasets for different cities or provinces.

In addition to these advantages, the approach has several practical applications:

1. Law Enforcement Agencies: Supports resource allocation and patrol planning by identifying high-risk areas

2. Government and Policy Makers: Assists in urban safety planning and policy decision-making

3. City Planning Authorities: Helps identify regions that require infrastructure or safety improvements

4. Public Safety and Research: Enables analysis of long-term crime patterns for research and strategic planning

Therefore, this project supports data-driven decision-making by combining pattern discovery with real-world applicability.

1.5. Structure of the Report

The remainder of this report is organized as follows:

Section 2 presents the background and related work, including relevant research and existing techniques in spatio-temporal crime analysis.

Section 3 describes the methodology and approach used in this project, including data processing, clustering, and hotspot evolution methods.

Section 4 explains the implementation details, including tools, system design, and key components of the developed system.

Section 5 presents the results obtained from the analysis, along with visualizations and interpretations of the findings.

Section 6 provides the conclusion, summarizing the main outcomes of the project.

Section 7 discusses the limitations and suggests directions for future work.

Section 8 lists the references used throughout the report.

2. Background

2.1. Previous work

In previous academic projects in University of Regina, most of the work focused on supervised machine learning tasks, for example, classification and regression (like Smart Lending: Personalized Loan Interest Rate Prediction), and computer vision (like Driver Drowsiness Detection System). These projects helped in building predictive models, performing feature engineering, and evaluating performance using standard metrics such as accuracy, MAE (Mean Absolute error) and MSE (Mean Squared Error).

Although, these approaches provided a strong foundation in data processing and model development, they mainly treated each data point independently and did not consider spatial or temporal relationships. Because of this, they were not suitable for analysing problems where location and time play a crucial role.

This project differs from previous work as it focuses on unsupervised learning and spatio-temporal analysis. This shift allows me to perform realistic analysis of real-world problems, where interpretability and long-term pattern understanding have more importance than short-term prediction.

2.2. Related Research

Several research studies have explored the problem of crime hotspot detection and spatio-temporal analysis using different analytical approaches.

A systematic review by [Butt et al. \[1\]](#) highlights that density-based clustering methods, particularly DBSCAN and Random Forest, are useful for identifying crime hotspots due to their ability to detect irregular cluster shapes and handle noise in terms of both accuracy and efficiency. However, the study also points out that DBSCAN is sensitive to parameter selection, which can affect clustering quality. Also, it needs to deal with sparsity, underlying population and demographics factors, etc.

[Ahmad et al. \[2\]](#) propose a method that combines kernel density estimation with machine learning models to detect and predict crime hotspots and named it CHART (Crime Hotspot Analysis and Real-time Tracking). Their approach generates heatmaps and supports real-time analysis. While this method provides strong predictive capabilities, it relies on labelled data and focuses more on prediction rather than pattern interpretation. Additionally, they were lacking in real-time data streams.

[Chen et al. \[3\]](#) apply a spatio-temporal clustering method called ST-DBSCAN to analyse crime patterns. Their work shows that incorporating both spatial and temporal dimensions can reveal meaningful clusters. However, the method consumes more time, and not efficient with the dataset having more noise.

A study on variable-density clustering for crime analysis by [Moews B. \[4\]](#) shows that crime hotspots often contain internal substructures and are not uniform in nature every time. This highlights the limitation of traditional methods that treat hotspots as simple regions. The study emphasizes the need for flexible and interpretable approaches to better understand spatial variations and support effective decision-making in crime analysis.

A study on crime pattern analysis using temporal modelling by [Li et al. \[5\]](#) shows that crime trends can be effectively analysed across different time periods, including seasonal variations. The results shows that temporal patterns in crime data are consistent and can support prediction models. However, the approach focuses more on prediction and statistical analysis rather than providing a clear framework for understanding spatial hotspot evolution.

All in all, existing research shows that clustering methods are effective for detecting crime hotspots, but many approaches either focus on prediction, increase model complexity, or lack a simple and interpretable way to track hotspot changes over time. This project addresses these limitations by using a clear and interpretable framework based on spatial clustering and centroid-based evolution tracking.

2.3. Existing Techniques

There are several common techniques used for crime hotspot detection and spatio-temporal analysis.

Density-based clustering methods, such as DBSCAN, are widely used for spatial hotspot detection. These methods are effective because they can identify clusters of arbitrary shape (i.e. random shape) and naturally handle noise in the data. However, it is sensitive to parameter selection, particularly the choice of distance threshold and minimum number of points required to form a cluster.

Kernel Density Estimation (KDE) is another commonly used technique for crime analysis. It generates smooth heatmaps that represent the intensity of crime in different areas. Although, KDE is useful for visualization, it does not produce clearly defined clusters, which makes it difficult to compare hotspots across time periods.

Temporal segmentation is used to divide data into time windows, allowing analysis of how patterns change over time. The choice of time window size is important, as smaller windows may introduce noise, while larger windows may hide important variations.

Spatio-temporal clustering methods, such as ST-DBSCAN, combine spatial and temporal information into a single clustering process. These methods can capture more complex patterns but increase computational complexity and reduce interpretability.

Overall, each technique has their own pros and cons. Density-based clustering provides clear and interpretable spatial hotspots, while temporal segmentation allows analysis of pattern changes over time. This project combines these approaches by applying spatial clustering

within defined time windows, enabling a balance between interpretability and analytical depth.

2.4. Summary of Techniques

The review of existing research and techniques shows that spatial clustering methods are effective for identifying crime hotspots, while temporal segmentation is useful for analysing how these patterns change over time. However, many existing approaches either focus heavily on prediction, increase model complexity, or lack a clear and interpretable way to track hotspot evolution.

In this project, a combination of density-based spatial clustering and temporal segmentation is used to address these limitations. DBSCAN is applied to identify spatial hotspots within each time window, and a centroid-based matching approach is used to track how these hotspots evolve across consecutive periods.

This combination provides a simple and interpretable framework for analysing both the spatial distribution of crime and its temporal evolution, while avoiding the complexity of fully integrated spatio-temporal models.

The techniques used in this project were learned from a combination of academic research papers, practical documentation and from the course CS-715 (Advanced Data Science & Machine Learning) provided by University of Regina. Key concepts such as DBSCAN clustering and spatio-temporal analysis were studied from the referenced research works and CS-715, while implementation details were supported by official library documentation and course materials.

3. Approach

3.1. Data Source

The analysis in this project is based on the Chicago Crime Dataset, which is publicly available through the [City of Chicago data portal \[6\]](#). This dataset contains detailed records of reported crime incidents spanning multiple years.

The dataset includes over 8 million records covering the period from 2001 to 2026. Each record represents a single crime incident and contains attributes such as date and time of occurrence, geographic location (latitude and longitude), and crime type.

For this project, only the relevant attributes required for spatio-temporal analysis are used. These include the date of the incident, geographic coordinates (latitude and longitude), and the year extracted from the timestamp. Additional attributes such as crime type are retained as optional features for future crime specific analysis.

Before performing the analysis, the dataset is cleaned by removing records with missing or invalid values in key fields such as date, latitude, and longitude. The date field is converted

into a standard datetime format, and additional temporal features such as year and quarter are derived to support temporal segmentation.

Although the dataset spans more than two decades, the main analysis focuses on a multi-year subset from 2018 to 2025 to ensure consistency and meaningful comparison across time periods. A detailed analysis is also performed for the year 2023 to study quarterly hotspot patterns.

The developed system, referred to as the **“ST-Hotspot Analyser”** which is designed to detect and analyse spatio-temporal crime hotspots.

3.2. Data Preprocessing

Before performing clustering and analysis, the raw crime dataset is preprocessed to ensure data quality and consistency.

The dataset is first loaded, and relevant attributes are selected, including date, crime type, latitude, longitude, and year. Records with missing or invalid values in the date or geographic coordinates are removed, as these fields are essential for spatio-temporal analysis.

The date field is converted into a standard datetime format to enable temporal operations. From this field, additional features such as year and quarter are extracted. The quarter is represented in the format “YYYYQn” (e.g., 2023Q1), which is later used for temporal segmentation and dashboard preparation.

After cleaning, the dataset is filtered to include only records from the selected analysis period (2018 to 2025). This ensures consistency across time and avoids issues caused by incomplete or partial data in earlier or recent years.

The final processed dataset contains only valid records with complete spatial and temporal information, making it suitable for clustering and further analysis.

3.3. Spatial Hotspot Detection

Spatial crime hotspots are identified using the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm. This method groups together points that are closely located in space while labelling isolated points as noise. DBSCAN is widely used for hotspot detection due to its ability to identify irregular cluster shapes and handle noise effectively. [\[1\]](#)

DBSCAN is well suited for crime data because it does not require the number of clusters to be specified in advance and can detect clusters of arbitrary shapes. This is important since crime hotspots do not follow regular geometric patterns. Additionally, DBSCAN naturally handles noise, allowing sporadic (i.e. irregular/occasional) or isolated crime incidents to be excluded from hotspot analysis.

In this project, DBSCAN is applied to geographic coordinates (latitude and longitude) using the haversine distance metric. This distance measure accounts for the curvature of the Earth, making it appropriate for spatial analysis using real-world geographic data.

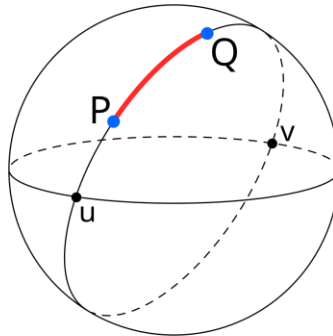


Figure 1: Illustration of the Haversine distance formula [7]

Before applying DBSCAN, the latitude and longitude values are converted into radians, as required for haversine distance calculations. The clustering is then performed separately for each temporal segment.

The key parameters used for DBSCAN are:

- **eps (epsilon):** 300 meters, which defines the maximum distance between points in the same cluster
- **min_samples:** 50, which defines the minimum number of points required to form a cluster

These parameter values are selected based on empirical testing and analysis of the k-distance plot. Smaller values of epsilon (e.g., 250 meters) resulted in fragmented clusters, while larger values (e.g., 400 meters) led to over-merged clusters. A value of 300 meters provided a balance between these extremes. The output of this step is a set of spatial clusters representing crime hotspots for each time window, along with noise points that do not belong to any cluster. Each cluster is further summarized using its centroid and size, which are used in subsequent analysis.

3.4. Temporal Segmentation

To analyse how crime patterns change over time, the dataset is divided into discrete temporal segments. In this project, the data is segmented into quarterly time windows (Q1–Q4) for each year.

Quarterly segmentation is chosen as a balance between temporal detail and data stability. Monthly segmentation can introduce excessive noise due to short-term fluctuations, while yearly aggregation may hide important variations in crime patterns. Using quarters provides enough data points within each segment while still capturing meaningful temporal changes.

For each year, the data is divided into four quarters in the format “YYYYQn” (e.g., 2023Q1, 2023Q2). Spatial clustering using DBSCAN is then applied independently to each quarter. This allows crime hotspots to form naturally within each time window without being influenced by data from other periods. This approach enables consistent comparison of hotspot locations across consecutive quarters, forming the basis for analysing hotspot evolution in the next stage.

3.5. Hotspot Evolution Method

To understand how crime hotspots change over time, clusters detected in consecutive quarters are compared using a centroid-based matching approach. For each cluster identified by DBSCAN, a centroid is calculated by taking the average latitude and longitude of all points within the cluster. These centroids represent the approximate location of each hotspot and are used as a basis for comparison across time periods.

To track hotspot evolution, centroids from one quarter are compared with centroids from the next quarter. The distance between centroids is computed using the haversine distance measure. If the distance between two centroids is less than or equal to a predefined threshold of 1.5 kilometres or 1 mile, the clusters are considered to represent the same hotspot. This threshold is selected to allow small spatial shifts in hotspot locations while still representing the same neighbourhood-level activity.

Based on this comparison, each hotspot is classified into one of the following categories:

- **Persisted:** A hotspot that continues to exist in the next quarter within the defined distance threshold
- **Disappeared:** A hotspot that does not have a matching cluster in the next quarter
- **Emerged:** A new hotspot that appears in the current quarter without a corresponding match in the previous quarter

This method provides a simple and interpretable way to track hotspot changes over time without requiring complex models. It allows the identification of stable crime areas as well as regions where crime patterns are evolving. However, this approach has certain limitations. Since it relies only on centroid distance, it does not capture the full spatial shape or structure of clusters. In addition, small changes in cluster boundaries may affect the matching results. Despite these limitations, the method offers a practical and effective solution for analysing spatio-temporal crime patterns in a large-scale dataset.

The computational performance of the proposed method is efficient for the given dataset. For example, clustering for a single quarterly segment can be completed in under 1 minutes on a system with an Intel Core Ultra 7 processor and 32 GB RAM. The use of spatial indexing (ball tree) improves the efficiency of the DBSCAN algorithm. Processing the data in temporal segments also helps manage memory usage and scalability for multi-year analysis.

The developed software processes crime data, performs clustering and hotspot evolution analysis, and generates structured outputs that are used for visualization in a Power BI dashboard. The system provides summary statistics, cluster information, and evolution labels as output.

3.6. Example Workflow

To illustrate how the proposed method works, a simplified example of the workflow is described below.

Input:

The input consists of crime incident records within a specific time window, for example, one quarter of a given year (e.g., 2023Q1). Each record includes the date, latitude, and longitude of the crime incident.

Example input:

Table 1: Sample crime data for workflow illustration

Date	Latitude	Longitude
2023-01-10	41.88	-87.63
2023-01-12	41.89	-87.64
2023-01-15	41.87	-87.62

Processing Steps:

1. The input data is preprocessed to remove invalid records and extract the required features.
2. DBSCAN clustering is applied to the geographic coordinates to identify spatial crime hotspots.
3. For each detected cluster, a centroid and cluster size are calculated.
4. The same process is repeated for the next quarter (e.g., 2023Q2).
5. Centroids from consecutive quarters are compared using haversine distance.
6. Based on the distance threshold, hotspots are classified as persisted, emerged, or disappeared.

Output:

The output includes:

- A set of clusters representing crime hotspots for each quarter
- Summary statistics such as number of clusters and noise percentage

- Evolution labels indicating how hotspots change between consecutive quarters

For example, if a cluster in 2023Q1 has a centroid close to a cluster in 2023Q2 (within 1.5 km), it is classified as a persisted hotspot. If no matching cluster is found, it is classified as disappeared. Any new cluster appearing in the next quarter is classified as emerged.

Figure 2 shows the overall workflow of the proposed spatio-temporal hotspot detection and evolution process.

This example demonstrates how the proposed pipeline transforms raw crime data into meaningful spatio-temporal insights in a structured and interpretable manner.

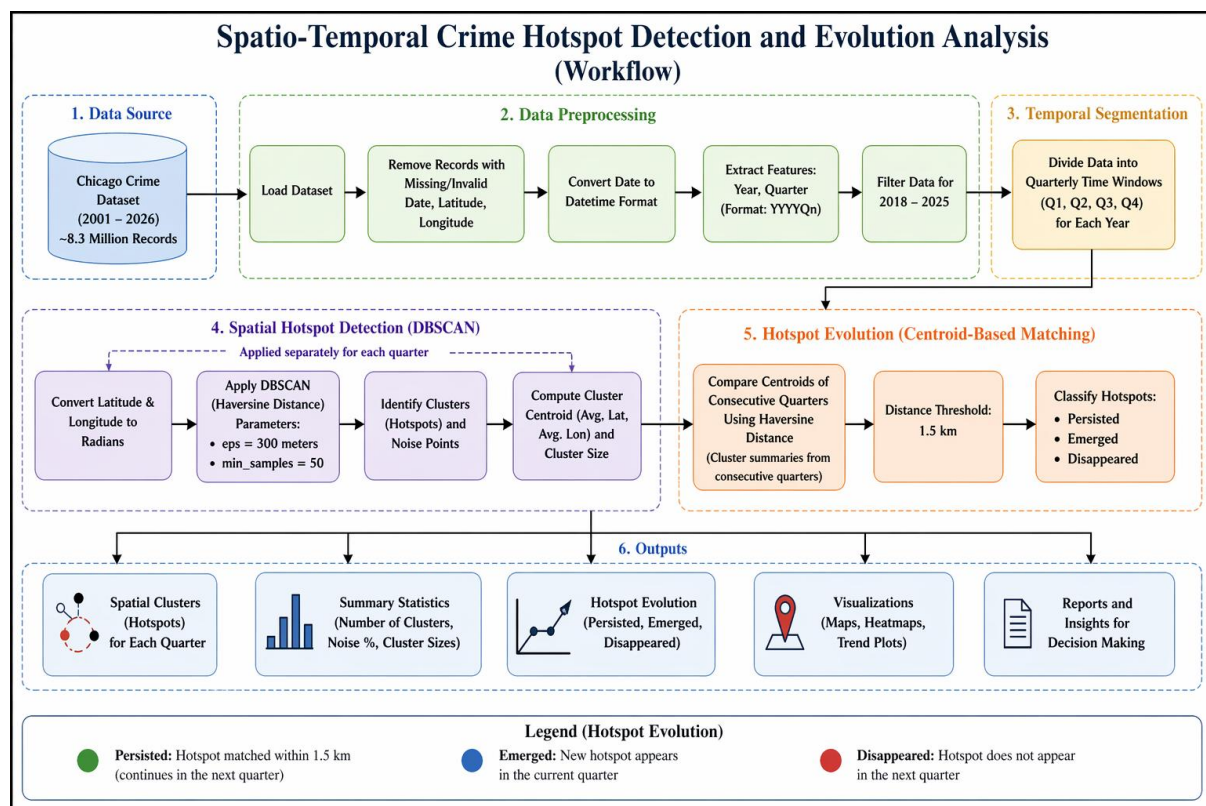


Figure 2: Workflow of the proposed ST-Hotspot Analyser for spatio-temporal crime hotspot detection and evolution analysis

4. Implementation

4.1. Tools and Technologies

The implementation of this project is carried out using a combination of programming tools and data analysis libraries.

Python is used as the primary programming language due to its strong support for data processing and machine learning tasks. The dataset is handled using the pandas library, which

provides efficient data manipulation and preprocessing capabilities. For numerical operations and array handling, NumPy is used.

The DBSCAN clustering algorithm is implemented using the scikit-learn library, which provides a reliable and optimized implementation of clustering techniques. Geospatial distance calculations are performed using the haversine formula, allowing accurate measurement of distances between geographic coordinates.

These tools were used to develop the ST-Hotspot Analyser, which performs clustering, hotspot evolution analysis, and generates structured outputs for visualization.

For data visualization and analysis, Power BI is used to create an interactive dashboard. The dashboard provides insights into spatial distribution, temporal trends, and hotspot evolution across different time periods. Additionally, Jupyter Notebook in VS Code is used for development and experimentation, enabling step-by-step execution and analysis of the workflow.

The implementation was carried out using Python 3.11 along with libraries such as pandas, NumPy, and scikit-learn, ensuring compatibility and stability during development.

Hardware Environment

The project was developed and executed on an Alienware 16X Aurora laptop with an Intel Core Ultra 7 processor, 32 GB DDR5 RAM and an NVIDIA GeForce RTX 5060 GPU. This hardware supported the processing of large crime datasets and helped run the preprocessing, clustering, and analysis workflows efficiently.

4.2. System Design

ST-Hotspot Analyser is designed as a modular pipeline that processes raw crime data and generates spatio-temporal insights in a structured manner. Each component performs a specific task, and the output of one stage serves as the input to the next.

The process begins with loading the cleaned crime dataset, which contains spatial and temporal information. The preprocessing module prepares the data by ensuring consistency and extracting required features such as year and quarter. The temporal segmentation module divides the dataset into quarterly subsets. Each subset is processed independently in the spatial clustering module, where the DBSCAN algorithm is applied to identify crime hotspots and noise points.

The clustering output is then passed to the centroid computation module, where each cluster is summarized using its average latitude, longitude, and size. These summaries are used by the hotspot evolution module, which compares clusters from consecutive quarters using distance-based matching. Based on this comparison, hotspots are labelled as persisted, emerged, or disappeared. The final outputs include cluster summaries, evolution results, and aggregated statistics.

These outputs are exported as structured data files and used in the visualization layer, where Power BI dashboards provide interactive exploration of spatial patterns, temporal trends, and hotspot evolution. This modular design ensures that each component can be independently analysed, tested, and improved, while maintaining a clear and interpretable overall workflow.

4.3. Key Code Snippets

1. DBSCAN Clustering

```
from sklearn.cluster import DBSCAN

import numpy as np

coords = np.radians(df_y[["Latitude", "Longitude"]].to_numpy())

earth_radius_m = 6371000

eps = 300 / earth_radius_m

db = DBSCAN (eps=eps, min_samples=50, metric="haversine", algorithm="ball_tree")

labels = db.fit_predict(coords)

df_y["cluster"] = labels
```

Explanation:

Geographic coordinates are converted to radians to support haversine distance. DBSCAN is applied to group nearby crime incidents into clusters while labelling isolated points as noise.

2. Cluster Centroid Calculation

```
cluster_summary = (
    df_y[df_y["cluster"] != -1]
    .groupby("cluster")
    .agg(
        size=("cluster", "count"),
        mean_lat=("Latitude", "mean"),
        mean_lon=("Longitude", "mean")
    )
    .reset_index()
)
```


Explanation:

For each cluster, the centroid is calculated using the average latitude and longitude. Cluster size is also computed for further analysis.

3. Hotspot Evolution Matching

```
def haversine_m(lat1, lon1, lat2, lon2):  
  
    R = 6371000  
  
    phi1, phi2 = np.radians(lat1), np.radians(lat2)  
  
    dphi = np.radians(lat2 - lat1)  
  
    dlmb = np.radians(lon2 - lon1)  
  
    a = np.sin(dphi/2)**2 + np.cos(phi1)*np.cos(phi2)*np.sin(dlmb/2)**2  
  
    return 2 * R * np.arcsin(np.sqrt(a))  
  
match_threshold_m = 1500  
  
if best_dist <= match_threshold_m:  
    status = "persisted"  
  
else:  
    status = "disappeared"
```

Explanation:

Distances between cluster centroids from consecutive quarters are calculated using the haversine formula. A threshold of 1.5 km or 1 mile is used to determine whether a hotspot persists. Clusters without a match are classified as disappeared, while new clusters in the next quarter are labelled as emerged.

5. Results**5.1. Dataset Overview**

The original Chicago Crime Dataset contained approximately 8.4 million crime records with 22 columns. After preprocessing, records with missing date, latitude, or longitude values were removed, and only the fields required for this project were retained. The final cleaned dataset contains approximately 8.3 million valid records from 2001 to 2026. For the multi-year analysis, the dataset was filtered to the period from 2018 to 2025, resulting in approximately 1.9 million records. The main fields used in the analysis were date, primary crime type, latitude, longitude, year, and quarter. These fields provided the required spatial and temporal information for clustering and hotspot evolution analysis.

Table 2: Dataset summary

Dataset Stage	Records	Description
Raw dataset	~8.4M	Original Chicago crime records
Cleaned dataset	~8.3M	Records with valid date and location
Multi-year subset	~2M	Records from 2018–2025

5.2. Clustering Results

Spatial clustering using DBSCAN is applied independently to each quarterly segment of the dataset. The algorithm identifies dense regions of crime incidents as hotspots while labelling isolated incidents as noise. For the detailed analysis, the year 2023 is considered. The dataset is divided into four quarters, and clustering is performed for each quarter separately. The results show that each quarter contains a significant number of clusters, indicating concentrated areas of crime activity. The number of detected clusters varies slightly across quarters, reflecting changes in crime distribution over time.

Table 3: Clustering results by quarter for 2023

Quarter	Records	Clusters	Noise (%)
2023 Q1	60,071	85	26.20%
2023 Q2	65,486	82	22.22%
2023 Q3	70,703	72	20.85%
2023 Q4	64,983	76	23.45%

These results indicate that the number of hotspots remains relatively stable across quarters, typically ranging between 70 and 85 clusters. The noise percentage varies between approximately 20% and 26%, showing that a portion of crime incidents do not belong to any dense region. Overall, the clustering results demonstrate that DBSCAN effectively identifies meaningful spatial patterns in crime data while filtering out scattered or isolated incidents.

5.3. Hotspot Evolution Results

To analyse how crime hotspots change over time, clusters from consecutive quarters are compared using centroid-based matching. Each hotspot is classified as persisted, emerged, or disappeared based on spatial proximity. The results for the year 2023 show that a large proportion of hotspots remain stable across consecutive quarters. This indicates strong spatial persistence in crime patterns, where certain geographic areas consistently experience higher crime activity.

Table 4: Hotspot evolution results across quarters for 2023

Transition	Persisted	Disappeared	Emerged	Persistence (%)
Q1 → Q2	58	27	24	53.21%
Q2 → Q3	58	24	14	60.42%
Q3 → Q4	52	20	24	54.17%

The persistence rate across quarters ranges between approximately 50% and 72%. This shows that most hotspots continue to exist in nearby locations over time, indicating stable crime regions. At the same time, a smaller number of hotspots either disappear or newly emerge in each quarter. These changes reflect shifts in crime patterns, which may be influenced by factors such as seasonal variation, local events, or changes in urban activity. Overall, the results demonstrate that crime hotspots exhibit both stability and variation. While many hotspots persist over time, new hotspots can emerge and existing ones can disappear, highlighting the dynamic nature of crime patterns.

5.4. Multi-Year Analysis

To understand long-term crime patterns, the analysis is extended across multiple years from 2018 to 2025. The same methodology is applied to each year, and summary statistics are computed to identify trends in hotspot behavior. The results show that the number of hotspots remains relatively consistent across years, with a gradual increase in some periods. The average number of clusters ranges from approximately 73 to 86 per year. The noise percentage varies across years, indicating changes in how crime incidents are distributed. Certain years exhibit higher noise levels, suggesting more scattered crime patterns, while others show more concentrated activity.

Table 5: Multi-year clustering and persistence summary (2018–2025)

Year	Average Clusters	Noise (%)	Persistence (%)
2018	77.00	23.22	59.72
2019	73.00	23.24	57.91
2020	73.00	32.74	49.83
2021	73.25	34.50	68.88
2022	81.25	28.35	72.82
2023	78.75	23.18	55.81
2024	86.00	23.60	63.14

2025	86.25	27.27	55.35
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5.5. Dashboard Visualization

To support interactive exploration of the results, a Power BI dashboard is developed. The dashboard provides a visual representation of spatial patterns, temporal trends, and hotspot evolution, making the analysis easier to interpret.

The dashboard consists of three main pages:

Overview Page

The overview page presents key performance indicators (KPIs) and overall trends. It includes metrics such as the average number of hotspots, persistence rate, noise percentage, and total number of crime records. A year-based filter allows users to explore trends across different time periods. Line charts show changes in hotspots, persistence, and noise over time, providing a high-level understanding of crime patterns. Figure 3 shows the overview dashboard.

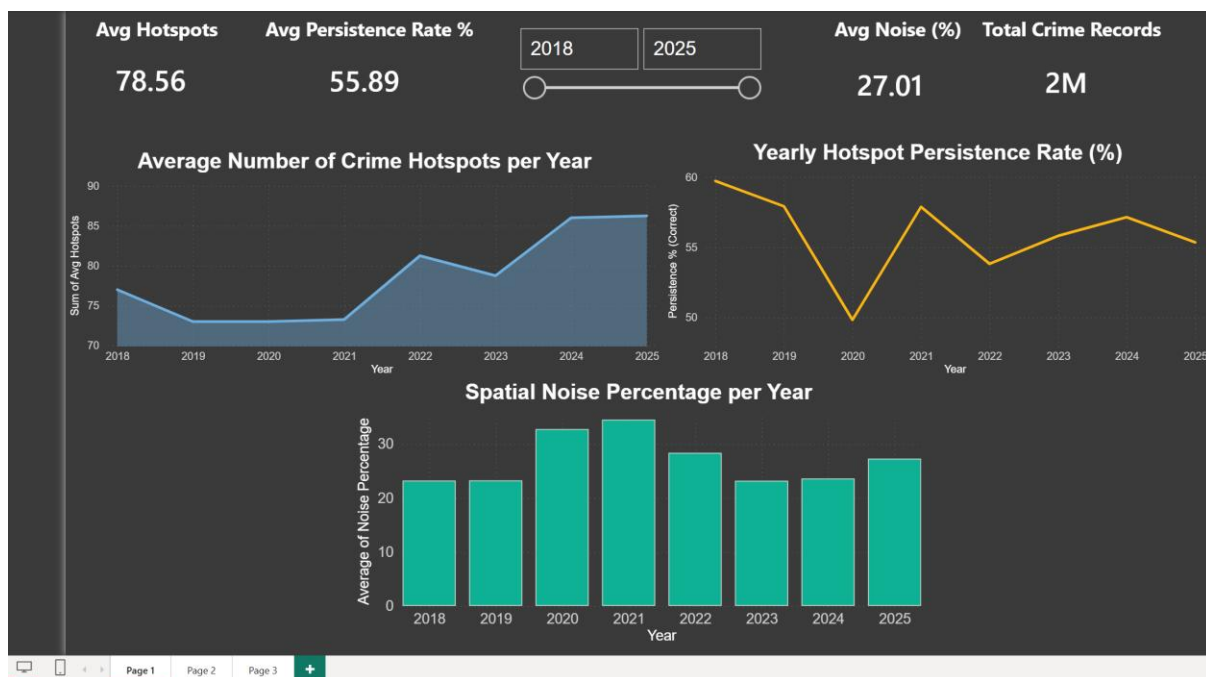


Figure 3: Overview dashboard with key metrics and trends

Quarterly Analysis Page

The quarterly analysis page focuses on variations within a selected year. It displays the number of hotspots and noise percentage for each quarter. This view helps identify seasonal patterns and short-term fluctuations in crime distribution. Users can select a specific year and observe how hotspot activity changes across quarters. Figure 4 shows the quarterly analysis dashboard.



Figure 4: Quarterly hotspot and noise analysis

Spatial Map Page

The spatial map page visualizes crime hotspots on a geographic map. Each cluster is represented as a bubble, where the size indicates the cluster size and the colour represents hotspot status: green for persisted, yellow for emerged, and red for disappeared. Filters allow selection of year and quarter transitions (e.g., Q1 → Q2), enabling users to observe how hotspots evolve spatially. Figure 5 shows the spatial hotspot map.

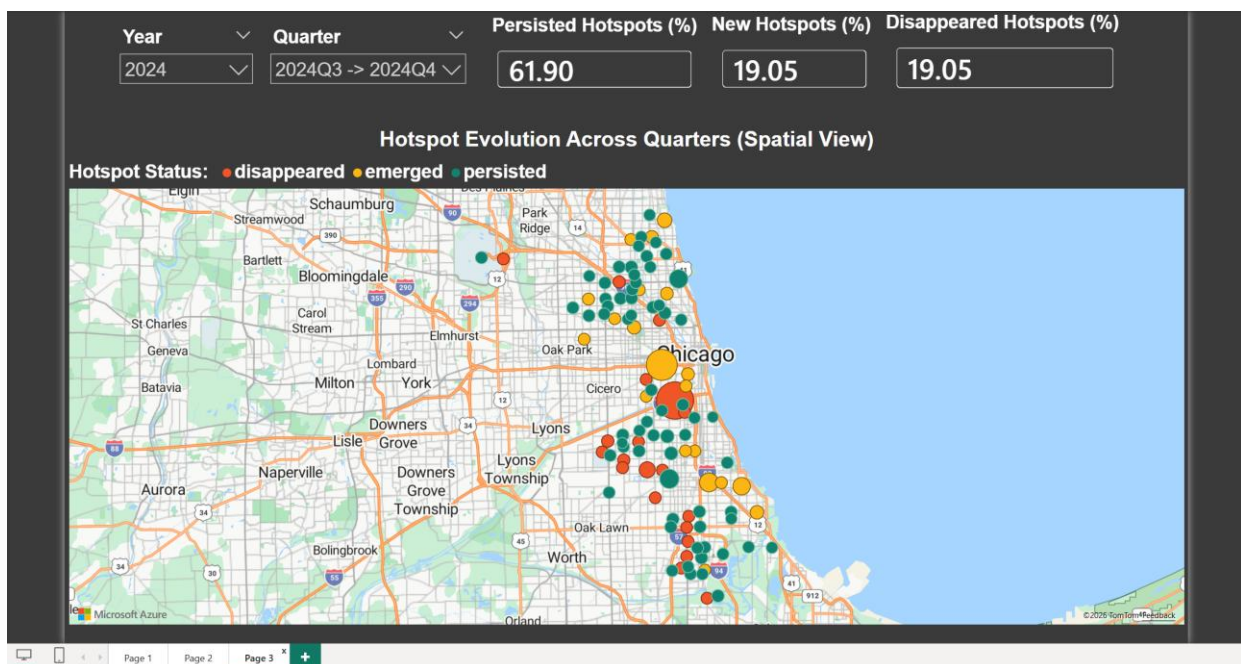


Figure 5: Spatial map showing crime hotspots and their evolution across consecutive quarters

Overall, the dashboard enhances the usability of the analysis by providing an interactive and intuitive way to explore complex spatio-temporal patterns.

6. Conclusion

This project presents a spatio-temporal analysis of urban crime data using an unsupervised learning approach. The main objective was to detect spatial crime hotspots and analyse how these hotspots evolve over time. The proposed method combines density-based spatial clustering with temporal segmentation to identify meaningful patterns in large-scale crime data. DBSCAN is used to detect hotspots within each quarter, and a centroid-based matching approach is applied to track hotspot evolution across consecutive time periods.

The results show that crime hotspots are generally stable, with a significant proportion persisting across quarters. At the same time, some hotspots emerge and disappear, indicating that crime patterns are both stable and dynamic. The multi-year analysis further confirms that these patterns remain consistent over longer periods, with moderate variations across different years. An interactive dashboard is developed to visualize the results, enabling intuitive exploration of spatial distribution, temporal trends, and hotspot evolution.

Overall, this project successfully addresses the problem of analysing spatio-temporal crime patterns and demonstrates that unsupervised learning techniques can effectively uncover meaningful insights from large-scale crime data. The findings can support data-driven decision-making in areas such as resource allocation, policy planning, and urban safety.

7. Future Work

While the proposed approach provides meaningful insights into spatio-temporal crime patterns, there are several areas for improvement and extension. One limitation of the current method is the use of centroid-based matching for tracking hotspot evolution. This approach considers only the central location of clusters and does not capture their full spatial shape. Future work can explore more advanced methods, such as cluster overlap or similarity-based matching, to better represent changes in hotspot structure.

Another limitation is that the analysis does not maintain long-term identities of clusters across multiple time periods. Extending the method to track hotspot lifecycles over longer durations could provide deeper insights into persistent and recurring crime regions. From an implementation perspective, processing very large datasets may increase computational time and memory usage. Future improvements can focus on optimizing performance and exploring scalable approaches for handling larger datasets or finer temporal resolutions.

The current implementation also focuses primarily on spatial and temporal attributes. Future extensions can incorporate additional features such as crime type, time of day, or seasonal effects to enable more detailed and context-aware analysis. In addition, machine learning models can be integrated to forecast future hotspot locations based on historical patterns,

extending the system toward predictive analysis. Overall, these improvements can enhance both the accuracy and practical applicability of the proposed approach.

8. References

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